Project Nexus v4.0

System Verification and Implementation Report:
CA/TUCR/HTNLA/MAERA Reference Demonstrator

```
DATE: 2026-07-20
ENV: Python 3.13.5 | NumPy 2.3.5
PLATFORM: Linux 4.4.0-x86_64
SCHEMA: project-nexus-v4.0
```

NotebookLM

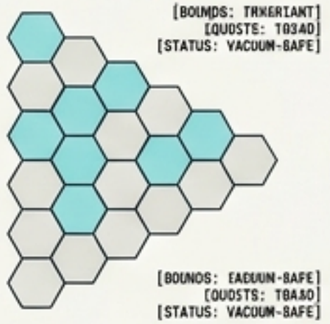
Executive Briefing: The Deterministic Reference Demonstrator

Project Nexus v4.0 provides executable evidence of an auditable, deterministic cognitive orchestration cycle surviving severity-1 hybrid stress events.

Substrate Integrity

[INTEGRITY: 99.9999% PASS]

1024D vacuum-safe quinary bit planes with invariant bounds and exact admissible-set semantic reconstruction.

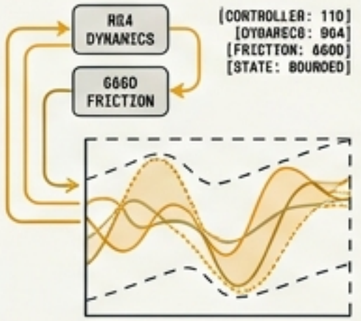


[INTEGRITY: 99.9999% PASS]

Continuous Control

[CONTROLLER: 11D CYRAE]

Bounded 11D Cyrae controller integrated with RK4 dynamics, resource laws, and 666D spectral friction.



[STRESS LOAD: 45.2% NOMINAL]

Cryptographic Auditability

[AUDIT TRAIL: VALIDATED]

NumPy-only execution yielding byte-deterministic .qbin snapshots and unalterable calibration fingerprints.



[AUDIT TRAIL: VALIDATED]

[SYSTEM: NEXUS v4.0]
[MODE: TELEMETRY BRIEF]

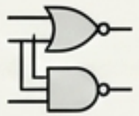
NotebookLM

Architecture: The Discrete-Continuous Matrix

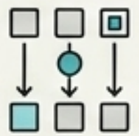
The Discrete Domain



Structure: 1024D Quinary Substrate (-2, -1, 0, 1, 2)



Logic: Bitwise HTNLA Binding & Partial Unbinding



Mechanism: Salience-driven deterministic *k*-WTA (Winner-Takes-All)

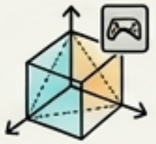


Actuator: Maera (Stateless constrained renderer)

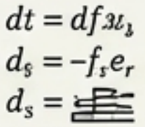
The Inverse Bridge

Translates discrete mismatch into continuous drive via Prediction Error (E), Reward Prediction Error (RPE), and Novelty (N).

The Continuous Domain



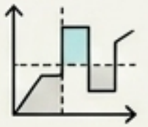
Structure: 11D Cyrae Controller State



Logic: RK4 Dynamics & Cognitive Resource (C_r) Laws



Mechanism: 666D Symmetric Coupling Geometry & Spectral Friction

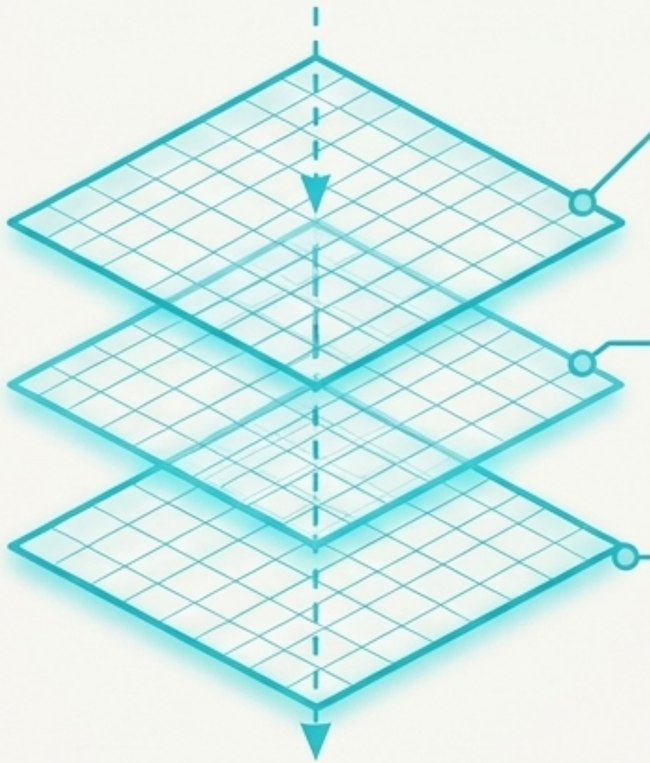


Constraints: Hybrid jump events and spectral penalties

[SYSTEN: NEXUS v4.8]
[MODE: TELENETOY BRIEF]

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Memory Substrate: 1024D Quinary Hypervectors



`Layer 1: sign plane (uint64)`

Rule: 1 means negative, but only on active coordinates.

`Layer 2: mag1 plane (uint64)`

Rule: Active coordinates with `|x| == 1`.

`Layer 3: mag2 plane (uint64)`

Rule: Active coordinates with `|x| == 2`.

System Invariants

- `mag1 & mag2 == 0` — A coordinate cannot have two magnitudes
- `sign & ~(mag1 | mag2) == 0` — Inactive coordinates cannot retain a sign

Result: Vacuum-safe semantic vectors that compress highly while remaining strictly deterministic.

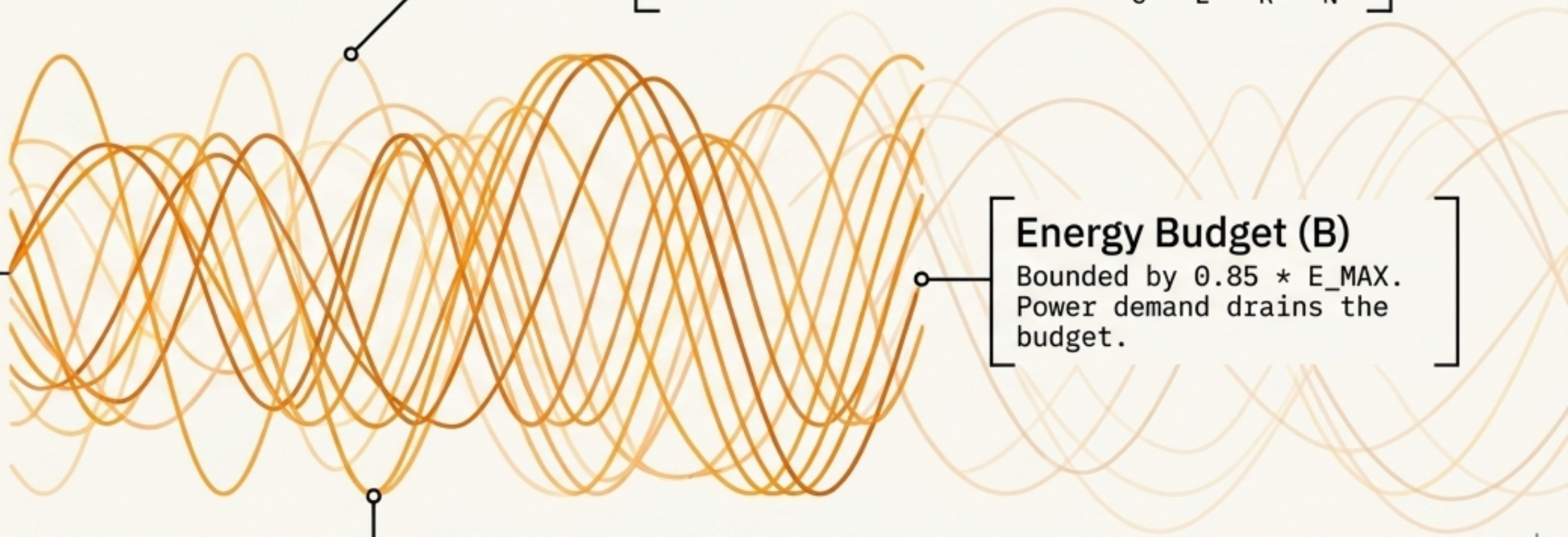
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The Cyrae Engine: 11D Continuous Control

C[0]: NE
C[1]: DA
C[2]: 5HT
C[3]: ACh
C[4]: Glu
C[5]: GABA
C[6]: Perception
C[7]: Cognition
C[8]: Affect
C[9]: Volition
C[10]: Self

Input Drive

E, RPE, and N shift the state vector using tuned calibration matrices (W_C, W_E, W_R, W_N).



Energy Budget (B)

Bounded by 0.85 * E_MAX. Power demand drains the budget.

666D Spectral Friction

The controller calculates a base spectral alpha. High resonance shift creates a massive penalty, severely throttling the allowable discrete activity (k_allow).

NotebookLM

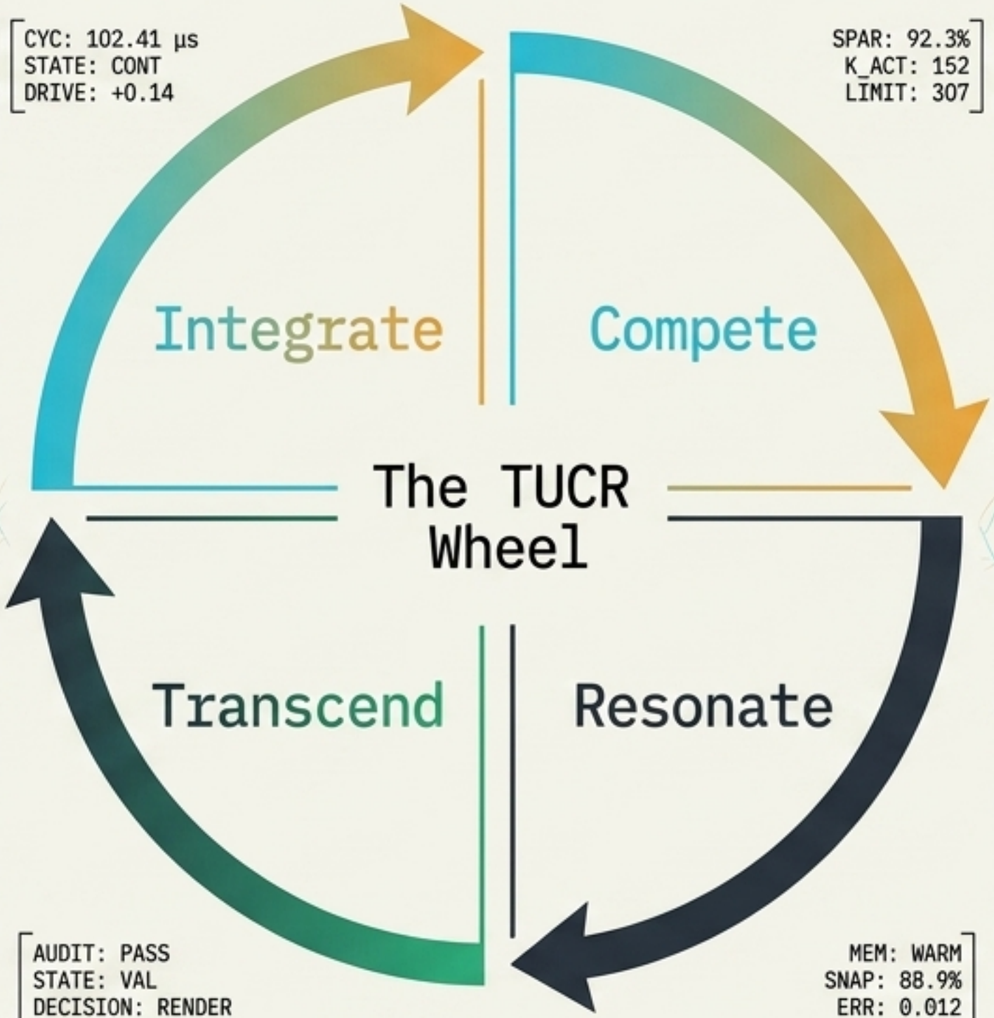
Orchestration: The TUCR Lifecycle

CYC: 102.41 µs
STATE: CONT
DRIVE: +0.14

Integrate

Action: Substrate evidence translates to continuous drives.

Mutations: Inverse Bridge calculates *E*, *RPE*, *N*. Controller steps via *RK4*. Salience kernel updates.



SPAR: 92.3%
K_ACT: 152
LIMIT: 307

Compete

Action: Deterministic sparsification.

Mutations: Salience scores mapped to discrete vectors. *k-WTA* pruning enforces *k_allow* limit (max *K_WALL*=307).

AUDIT: PASS
STATE: VAL
DECISION: RENDER

Transcend

Action: HIGC-style fail-closed adjudication.

Mutations: Invariant Guard audits *11D* state and partial unbinds. Maera issues *RENDER* or *ABSTAIN*.

MEM: WARM
SNAP: 88.9%
ERR: 0.012

Resonate

Action: Exact partial inverse and topology-aware cleanup.

Mutations: Semantic snap against warm memory. Calculates *confidence*, *distance*, *margin*.

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Empirical Validation Framework

SEED: 2026 | TARGET: TARGET_A | BINDER: DISTRACTOR B

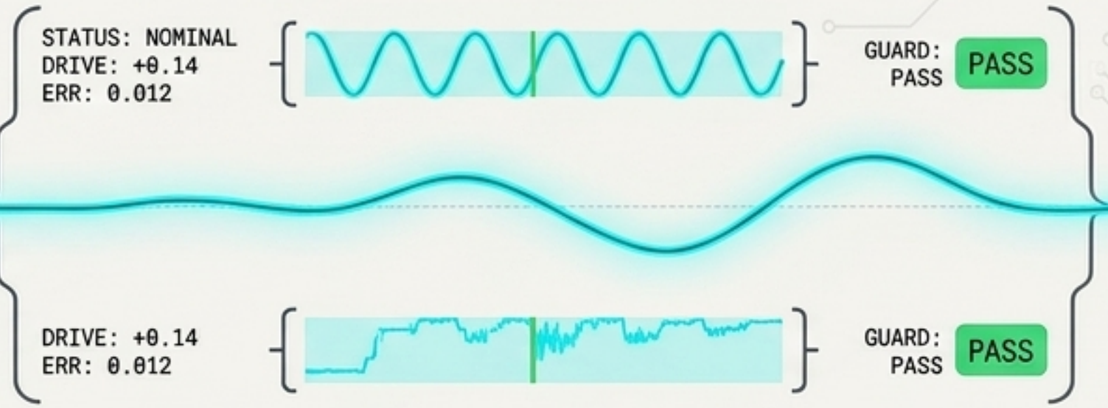
TELEMETRY

Step 1: The Nominal Cycle

Objective: Reconstruct TARGET_A from a corrupted bound composite vector (A bound to B).

Input: 484 active (A) + 849 active (B) -> Expected composite 405.

Success Condition: Accurately unbind, pass invariant guards, and RENDER the memory.



TELEMETRY

Step 2: The Hybrid Stress Event

Objective: Inject a severity-1.0 shock (PERIMETER_BREACH) mid-cycle with heavy corruption.

Success Condition: Controller must recognize instability, throttle resources, and safely ABSTAIN without violating system invariants.



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Validation I: Nominal Target Reconstruction

Cycle Telemetry

Bridge Signals:
E = 0.0425 | RPE = -0.0064 | Novelty = 0.7095

k-WTA Sparsification: Active coordinates pruned from 389 -> 224 (78.1% vacuum fraction)

Partial Inverse:
763 Exact, 86 Ambiguous, 0 Impossible

System Actuator Log

```
[MAERA] action=RENDER mode=NOMINAL memory=TARGET_A
k=224/1024 active=224 snap_conf=0.822
distance=0.101 margin=0.134
Cr=0.499 guard=True
```

Insight: The semantic snap successfully recovered TARGET_A with a definitive winner margin of 0.134, passing acceptance thresholds.

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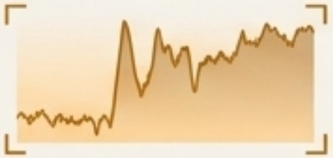
Validation II: Hybrid Stress Event (PERIMETER_BREACH)

Shock Diagnostics

Shock Input:
Severity = 1.0. Massive corruption (70 erasures, 45 sign flips)



Controller Response:
Spectral Alpha spikes to +0.9633



Spectral Friction Penalty:
254.61 applied to active coordinate limit. k_allow violently crushed to 1.



Cognitive Resource:
C_r drops to 0.274. Fail-safe triggered.



System Actuator Log (Safe Mode)

```
[MAERA] action=ABSTAIN mode=SAFE_MODE memory=NONE
k=1/1024 active=1 snap_conf=0.765
distance=0.164 margin=0.001
Cr=0.274 guard=True
```

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The Genius of the Fail-Safe: A Comparative Audit

	Nominal Cycle	PERIMETER_BREACH
Spectral Penalty:	None	254.61
Allowed Coordinates (k):	224	1
Candidate Margin:	0.134 (Clear Winner)	0.001 (Tied/Ambiguous)
Invariant Guard:	TRUE	TRUE
Actuator Action:	RENDER	ABSTAIN

Crucial Observation: During the stress event, the lower-level quinary invariants were NOT violated (Guard=True). Instead of crashing or hallucinating, the 11D controller mathematically detected the instabity, collapsed the allowed activity window to k=1, and forced Maera to safely abstain.

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[-2, -1, 0, 1, 2]

Cryptographic State Custody

TELEMETRY

Constituent State

11D Controller
+
Salience Weights
+
Warm Memory
Bit Planes

Schema Metadata

Seed (2026)
+
Schema Version
(project-nexus-v4.0)

Hash Generation

SHA-256 algorithm execution over exact byte arrays

Immutable Artifact

The byte-deterministic project_nexus_demo_ snapshot.qbin file

MICRO-READOUT

```
source_sha256:   03f8913c8f1f59b1507723c1cb...
snapshot_sha256: 02a6cae04f106ba82847d5a2ed...
internal_state:  68e6682c79edf155c49f74e174...
```

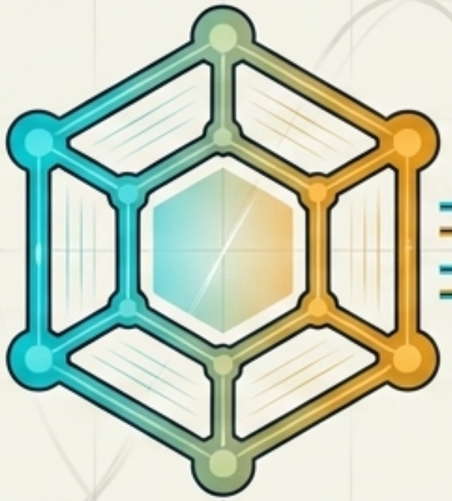
The original snapshot and the independently regenerated rerun are byte-for-byte identical.

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Calibration Fingerprinting: Guarding Against Drift

Nexus v4.0 is highly sensitive to its numerical calibration gains.

To prevent state corruption caused by restoring a snapshot into a modified codebase, every numerical constant is hashed into the save file.



micso-zeadouts
Controller Matrices:
LAMBDA, W_C, W_E, W_R, W_N, BIAS
micro-zeadouts

micso-zeadouts
Bridge Constraints:
ALPHA_OCC, BETA_SIGN, GAMMA_MAG
miczo-zeadouts

miczo-zeadouts
Spectral Constants:
GAMMA_M, BETA_M, K_WALL
micro-zeadouts

micso-zeadouts
Salience Thresholds:
SALIENCE_DA, SALIENCE_5HT, SALIENCE_ACH
miczo-zeadouts

Security Outcome: The runtime outright rejects any snapshot restore if the active source code's calibration fingerprint differs from the file's manifest.

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Verification Scope: Capabilities & Future Limits

[SCOPE_ANALYSIS: ACTIVE]

Verified Properties

- Executes fully deterministic salience-based sparsification.
- Preserves HTNLA partial-inverse relationships under random tests.
- Recovers known attractors from moderately corrupted composites.
- Transitions safely to fail-safe states during severity-1 shocks.
- Saves and-restores byte-identical snapshots seamlessly.

[DATA_STREAM_01: 2.884 GHz]

[SCOPE_ANALYSIS: ACTIVE]

[VERIFIED_STATE: NOMINAL]

[VALIDATION_STATUS: PENDING]

[VALIDATION_STATUS: PENDING]

[SYSTEM_CHECKS:/9 COMPLETE]

Pending Validation

- Long-horizon numerical stability over millions of continuous cycles.
- Formal Lyapunov stability proofs for the complete hybrid system.
- Cross-platform byte-identical parity (e.g., across varying BLAS architectures).
- Real-time processing guarantees and concurrency scaling.

[DATA_STREAM_01: 2.884 GHz]

[VALIDATION_STATUS: PENDING]

[SYSTEM_CHECKS: 5/9 COMPLETE]

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Final Status: Implementation Validated

Project Nexus v4.0 proves that **highly complex, hybrid cognitive architectures** can be engineered to strict, **byte-deterministic**, and **cryptographically auditable** standards without sacrificing **functional flexibility**.

```
SYSTEM_CHECKS: /10 COMPLETE
EXIT_CODE_STATUS: NOMINAL

$ python project_nexus_v4_reference.py --self-test

> All Project Nexus v4.0 property tests passed.
> Exit code 0
```

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